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**BIT 2324 GEOGRAPHICAL INFORMATION
SYSTEMS**

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This presentation is intended to be covered within one week. The notes, examples and exercises should be supplemented with a good textbook. Most of the exercises have solutions/answers appearing elsewhere and accessible by clicking the green **Exercise** tag. To move back to the same page click the same tag appearing at the end of the solution/answer.

Errors and omissions in these notes are entirely the responsibility of the author who should only be contacted through the **Department of Curricula & Delivery (SODeL)** and suggested corrections may be e-mailed to elarning@jkuat.ac.ke



LESSON 4

Tabular Data Design

Learning outcomes

By the end of this topic you should be able to;

- *Define data modeling.*



4.1. Introduction

Tables serve as a container for records about entities that can be distinguished by their attributes. Data modeling is the process of defining features to be included in the database, their attributes and relationships, and their internal representation in the database. It involves the development of conceptual, logical and physical models of the census geographic database. The outcomes include a comprehensive data dictionary that defines the content of the databases that are produced by the agency. In some instances, such dictionaries can be adopted or adapted from other agencies in the country, for example, where a national digital topographic database exists. In other cases, this data dictionary has to be developed from scratch. The resources required for this will depend on how comprehensive the database



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will be. It may also be necessary to integrate existing database models that have been developed to manage tabular census information. This is necessary, for example, if data from previous censuses need to be integrated in the GIS databases. In addition to the data dictionary, a procedural manual defines the steps necessary in the development and processing of digital spatial data. Such manuals are important for ensuring consistency in the products that are generated by different technicians or units that may, perhaps, be scattered across the country. They also define recurring analytical work such as the methods used to reconcile past census data with new boundaries after the administrative units have been changed. Accuracy standards should also be defined as part of the overall database design process. While accuracy is often not a critical issue in census mapping –



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in fact, many countries rely on hand-drawn sketch maps for this purpose – it becomes an issue when the resulting census maps are used in combination with other, higher-accuracy data.

Several types of database models exist, such as the flat, hierarchical, network, and relational models. A flat database is essentially a spreadsheet whereby all data are stored in a single, large table. A hierarchical database is also a fairly simple model that organizes data into a “one-to-many” association across levels. Common examples of this model include phylogenetic trees for classification of plants and animals and familial genealogical trees showing parent-child relationships. Network databases are similar to hierarchical databases, however, because they also support “many-to-many” relationships. This expanded capability allows greater search flexibility within the dataset and re-



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duces potential redundancy of information. Alternatively, both the hierarchical and network models can become incredibly complex depending on the size of the databases and the number of interactions between the data points. Modern geographic information system (GIS) software typically employs a fourth model referred to as a relational database.

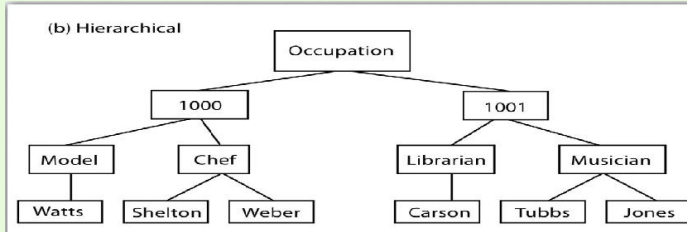


(a) Flat File

Name	Group #	Occupation
Watts	1000	Model
Shelton	1000	Chef
Weber	1000	Chef
Tubbs	1001	Musician
Jones	1001	Musician
Carson	1001	Librarian

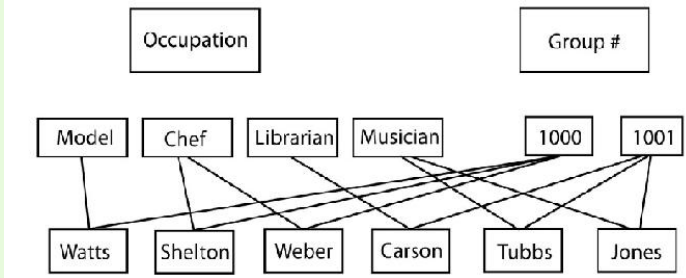


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(c) Network



4.2. Querying GIS data

1. Attribute query

- Select features using attribute data (e.g. using SQL)
- Results can be mapped or presented in conventional database form



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- Can be used to produce maps of subsets of the data or choropleth maps
2. Spatial query is used by clicking on features on the map to find out their attribute values. The combination of these is a powerful way of exploring spatial patterns in your data.



Revision Questions

EXERCISE 1.  What is data modeling?

Example . Discuss the two types of queries.

Solution: Revise.

EXERCISE 2.  Distinguish the different types of database files.



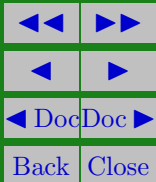
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Solutions to Exercises

Exercise 1. Revise.

Exercise 1





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Exercise 2. Revise.

Exercise 2

